

NATIONAL INSTITUTE OF TECHNOLOGY, TRICHY-620 015
DEPARTMENT OF MATHEMATICS
ASSESSMENT -II

Code/Subject: MAIR 12/Linear Algebra & Calculus
Branch: I Semester B.Tech. (ICE-B)

Maximum Marks: 20
Time: 3.00-4.00 p.m.
Date: 28.11.2025

- ✓ 1. Let $\{x_n\}$ be a sequence of positive numbers that converges to $\frac{1}{2}$. Determine whether the given series is absolutely convergent

(a) $\sum_{n=1}^{\infty} \frac{x_n^n \cos n\pi}{n}$ (2.5)

(b) $\sum_{n=1}^{\infty} \frac{(-1)^n n!}{n^n x_1 x_2 \dots x_n}$ (2.5)

- ✓ 2. Which of the following series converge absolutely, which converge, and which diverge? Give reasons for your answers.

(a) $\sum_{n=1}^{\infty} \frac{(-1)^n (n+1)^n}{(2n)^n}$ (2.5)

(b) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} (n!)^2}{(2n)!}$ (2.5)

- ✓ 3. Derive the Jacobian of Cylindrical co-ordinates. (3)

- ✓ 4. Find the Taylor's series expansion of $\tan^{-1}\left(\frac{y}{x}\right)$ in powers of $(x-1)$ and $(y-1)$ upto the second degree terms. (3)

- ✓ 5. Identify and discuss the stationary points of $f(x, y) = x^4 + 2y^2 - 2x^2 - y^4$ and hence find the values, if any. (4)
